

Andrés Casillas García de Presno

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EDUCATION

- 2023 – 2025 **Rheinische Friedrich-Wilhelms-Universität Bonn, Germany**, *M.Sc. Mathematics with minor in Computer Science* (Final Grade: 9.8)
Thesis: *Machine Learning-Based Control Methods for an Optimal AMG Setup Strategy with Massive Distributed Parallelism*, defended with highest grade.
- 2019 – 2023 **National Autonomous University of Mexico**, *B.Sc. Mathematics* (Final Grade: 9.9)
Thesis: *A Number-Theoretical Approach to Cellular Automata*, defended with honors.

AWARDS & HONORS

- 2026 **NAFEMS Award** for the second-best graduate thesis in Germany, Switzerland, and Austria.
- 2025 **Sotero Prieto Medal** for the best bachelor's thesis in mathematics in Mexico.
- 2024 **Gabino Barreda Diploma** for the second-best class GPA.
- 2022-2024 **Academic Excellence Scholarship** from UNAM's Mathematical Institute (IMATE).
- 2020-2022 **TELMEX-Telcel Scholarship** for outstanding students.

WORK EXPERIENCE

Fraunhofer Institute for Algorithms and Scientific Computing, Bonn, Germany Sept 2024 – present

Machine Learning Scientist: Used several Machine Learning techniques to speed up algorithms for solving large sparse systems of equations arising in mathematical models and simulations. Improved average routine's performance by 41% over default settings.

National Autonomous University of Mexico - Faculty of Science, CDMX, Mexico Sept 2021 – June 2023

Teaching Assistant: Taught Linear Algebra I, Higher Algebra I-II, and Calculus I-IV for students enrolled in mathematics, computer science, and actuarial science. Learned to organize and transmit complex ideas clearly.

PROJECTS

Parameter Importance Analysis (Python): Executed a workflow training several ML models on a large matrix dataset, selected the best learner, and predicted optimal parameter subsets for fast solvers.

Lovász Theta Number Calculator (Julia, Jupyter): Julia-based tool approximating Lovász Theta Number for graphs using combinatorics, semidefinite programming, Monte-Carlo, and optimization.

Poisson Problem Galerkin Method (Python, Jupyter): Solved weak Poisson problem on unit square with zero boundary using Galerkin's method, including plots and error computations.

SKILLS

Technical Python, Julia, Jupyter, Bash, Fortran, C++, HTML, GitHub, Latex, Linux, Copilot.

Languages Spanish (native), English (C2), German (B1), French (A2).

PUBLICATIONS

Casillas-García de Presno, A., Godínez, F. (2022). Construction of empirical models via stepwise fitting of a fractional Newtonian cooling law. *Fractals*, 30(3), 2250054. <https://doi.org/10.1142/S0218348X22501225>